

DIALLO BOYE

Senior Data Scientist | ML & Forecasting

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PROFILE

Senior data scientist with 6+ years at CERN and Brookhaven National Laboratory, specialising in statistical modeling, time series analysis, and end-to-end ML pipeline development on large-scale, high-dimensional datasets. Experienced in owning projects from problem framing to production, translating complex quantitative findings into actionable insights, and leading cross-functional teams. Author of multiple peer-reviewed publications in data-driven analysis and predictive modeling.

CORE COMPETENCIES

Time Series & Forecasting Temporal modeling, trend/seasonality decomposition, backtesting, cross-validation	Machine Learning Classification, regression, XGBoost/LightGBM, neural networks, feature engineering	Statistical Modeling Bayesian & frequentist inference, uncertainty quantification, experimental design, Monte Carlo simulation & optimization
Data Engineering Petabyte-scale pipelines, distributed computing, real-time processing, SQL	Programming Python (NumPy, pandas, scikit-learn, SciPy, PyTorch), C++, SQL, Linux/bash, Git, CI/CD, Docker	Communication & Leadership Cross-functional team lead, stakeholder reporting, 6+ international conference talks

PROFESSIONAL EXPERIENCE

Goldhaber Distinguished Fellow - Brookhaven National Laboratory / CERN 2023 – 2026

Highly competitive fellowship awarded to fewer than 5 early-career scientists per year worldwide.

- ▶ Developed time series analysis pipelines for beam parameter evolution — extracting trends, modeling non-linear dynamics, and delivering sub-percent precision results used across an international collaboration of 3,000+ scientists.
- ▶ Led a cross-functional team of ~20 researchers, setting technical direction, coordinating deliverables, and mentoring junior scientists across multiple concurrent projects.
- ▶ Designed and deployed automated ML pipelines replacing thousands of hand-tuned fits with a unified, scalable model — improving robustness and precision in production.
- ▶ Invited plenary speaker at LHCP 2026 (Paris) — selected to present a cross-experiment review to 500+ scientists; 6 further international conference talks.

Postdoctoral Research Associate - Brookhaven National Laboratory 2020 – 2023

- ▶ Built end-to-end ML pipelines on petabyte-scale datasets: feature engineering, model training, cross-validation, backtesting, and production monitoring — with results published in peer-reviewed journals.
- ▶ Applied multivariate classifiers and statistical inference to extract rare signals from overwhelming backgrounds — directly analogous to demand forecasting in volatile, noisy environments.
- ▶ Mentored 4 graduate students across 3 continents in quantitative methods, Python, and scientific software development.

PhD Researcher - CERN / ATLAS 2016 – 2020

- ▶ Implemented full data science workflows: simulation, classification, anomaly detection, hypothesis testing, and automated reporting on multi-terabyte datasets.

PROJECTS

swiss-power-forecast - Probabilistic Electricity Price Forecasting github.com/dialloboye/swiss-power-forecast

- ▶ Swiss day-ahead price forecasting (ENTSO-E, 30k hourly obs.): LightGBM beats the naive baseline by 29% MAE under walk-forward backtesting; overconfident prediction intervals (56.5% vs 80% nominal coverage) diagnosed and recalibrated to 79.1% via conformalized quantile regression. Leak-free features enforced by unit tests; CI/CD.

market-regime-scan - Multiple Testing & False Discovery Control github.com/dialloboye/market-regime-scan

- ▶ Scan of 26 years of S&P 500 volatility: 27 local “3 σ ” regime changes reduce to 1 (COVID) under a realistic block-bootstrap null; look-elsewhere effect quantified with a three-null significance ladder (18 $\rightarrow$ 5 $\rightarrow$ 1 survivors) and toy-based global thresholds.

EDUCATION

PhD, Particle Physics - ATLAS / CERN / University of South Africa 2020

MSc, Physics of Energy & Energy Transition - Universite Paul Sabatier, Toulouse 2016

MSc, Subatomic Physics & Astroparticle - Universite de Strasbourg 2014

LANGUAGES French Native English Fluent